SOURAV ROY

Quantitative Research | Risk Analytics | Data Science

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GitHub | Portfolio | arXiv: 2508.19582

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PROFILE

I hold two separate master's degrees: an M.Sc. in Mathematics from TIFR-CAM, Bengaluru, and an M.Sc. in Computer and System Sciences from TIFR STCS, Mumbai. My research covers probability, randomized algorithms, convex geometry and optimization. I translate mathematical questions into reproducible computational experiments and apply the same evidence-led approach to financial time series, forecasting, portfolio evaluation, large transaction datasets and fraud-risk analysis. My work emphasizes assumptions, chronological validation, leakage control, numerical checks, reproducibility, benchmark comparison and clear limits on what results establish. Seeking quantitative research, data science and risk analytics roles.

TECHNICAL SKILLS

Programming & data: Python, C++17, NumPy, pandas, SciPy, scikit-learn, CVXPY, Clarabel, SQL, SQLite; C, MATLAB, SageMath, Mathematica.

Quantitative & statistical methods: Probability, PCA, Ridge regression, Random Forest, KMeans, Gaussian HMMs, covariance estimation, minimum-variance optimization, risk forecasting, convex optimization, randomized algorithms and numerical simulation.

Research & evaluation: Chronological validation, fixed-cutoff forecasting, backtesting with transaction costs, leakage auditing, forecast freezing, sampling, complexity analysis, numerical invariants, reproducibility checks, Python unittest and pytest.

Tools: Git, GitHub, JupyterLab / Jupyter Notebook, VS Code, conda / Anaconda, macOS Terminal, DB Browser for SQLite, sqlite3 CLI, Homebrew, Apple Clang (C++17), LaTeX. These reflect tools I have used, with different levels of proficiency.

QUANTITATIVE FINANCE, DATA & RISK PROJECTS

Quant AI Market Lab | Independent quantitative research | completed v1.0.0 | 2026 | **Repository**

- Built a validated eight-ETF workflow covering returns, covariance, PCA, KMeans and Gaussian HMM market-condition models, Ridge and Random Forest forecasts of next-five-day SPY return RMS, and long-only minimum-variance portfolios.
- Verified return identities, covariance symmetry/PSD properties, eigendecomposition and explained variance. The first three PCs explained 79.01% of standardized historical return variation; on 565 common chronological dates Ridge MAE was 0.003201 versus 0.003443 for Random Forest.
- Compared five covariance methods, three transaction-cost assumptions, nine benchmark scenarios and 32 strategy-period robustness checks. Minimum-variance approaches reduced realized volatility in reported subperiods but underperformed SPY buy-and-hold on raw return; May-August 2026 are described as pseudo-out-of-sample rather than an untouched holdout.

Quant Market Lab II - Signal Processing & Prospective Forecasting | v0.1 baseline frozen; extensions in progress | 2026 | **Repository**

- Fixed the information cutoff at 31 August 2026; implemented Ridge, constant and rolling 20-day RMS baselines at four lead times, preserved 12 versioned September-December forecasts, and recorded model/data/source fingerprints with target-timing and prefix-invariance tests.

- Across 87 historical zero-gap windows Ridge had the lowest average absolute error in the repository comparison. September remains a retrospective check; nine October-December five-session forecasts were published from the frozen August-cutoff CSV without refitting. I implemented EWMA with separately frozen August/October information sets, then reproduced signed, squared and absolute-return autocorrelation without changing the forecasts. November and December outcomes remain pending; autoregressive extensions remain future work.

PaymentGuard | Payment-fraud risk study | first baseline milestone complete | 2026 | **Repository**

- Checksum-verified and processed 1,090,000 synthetic AMLNet v2 transactions (729 MiB); audited 17 top-level columns and 47 nested metadata paths, detected non-chronological file order and target-revealing metadata covering all 1,411 fraud labels, defined a 13-predictor policy, prepared chronological splits, verified the SQLite import and explored training data with SQL.
- On February 2026 validation (167,990 transactions; 202 positives), the log-amount logistic baseline caught 196 fraud labels at a predeclared 1% review budget (97.03% recall, 11.67% precision), versus 188 for raw-amount logistic regression and 109 for largest-amount ranking. The expanded checkpoint reports 72 passing tests. Feature ablations, score reconstructions and hypothetical review-cost and missed-case diagnostics are complete. Calibration, drift, final-period evaluation and a live decision policy remain open; synthetic-data scores do not establish real-payment performance.

ResearchAudit | Evidence-based validation project | rule-based baseline published | 2026 | **Repository**

- Defined an evidence-based multi-agent audit plan and froze Quant AI Market Lab v1.0.0 as the first target. I published an AST-based screening baseline: 34 Python files, 10 triaged findings and 52 passing test methods. Six claims retain provisional inspection support. Claim 7 is supported for tested cases with two input-validation gaps, recorded by characterization tests. Full numerical reproduction, independent detection-accuracy evaluation and multi-agent experiments remain open.

MATHEMATICAL & COMPUTATIONAL RESEARCH

Approximating Mixed Volumes to Arbitrary Accuracy | 2025 | Hariharan Narayanan and Sourav Roy | **arXiv paper**

- Co-authored a randomized approximation algorithm for mixed volumes of convex polytopes with integer vertices, combining convex optimization, Lorentzian polynomials and polytope subdivision; established multiplicative-error and failure-probability guarantees with polynomial runtime for a fixed number of distinct input polytopes under the stated assumptions.

Master's Thesis: Approximation of Mixed Volumes of Special Convex Bodies | TIFR Mumbai | March 2026 | Supervisor: Prof. Hariharan Narayanan | **Thesis PDF**

- Studied mixed volumes of a polytope and Euclidean ball using Minkowski sums, the Steiner polynomial, geometric decomposition, sampling and log-concavity; developed and analysed noisy binary search for a scaling parameter, including decision-error and runtime bounds, while distinguishing the idealized oracle model from practical approximate optimization.
- Implemented random-walk samplers for estimating volumes of convex bodies, including ball walk and hit-and-run variants, together with geometric membership/decomposition procedures and mixed-volume computations.

Exploring the Efficient Computation of Permanents of Graph Laplacians | TIFR | December 2022 | Supervisor: Prof. Piyush Srivastava | **Project report PDF**

- Proved the permanent analogue of Brenner's result for diagonally dominant matrices; explored tensor-product formulations, recursive expressions for paths, cycles and trees, and Pfaffian methods.

RSA Encryption and Decryption in Python | INSPIRE Summer Project, Presidency University Kolkata | 2019; revised 2026 | **Repository**

- Implemented educational RSA key generation and encryption/decryption using Miller-Rabin primality testing, the extended Euclidean algorithm and modular exponentiation. The 2026 revision adds UTF-8/ input validation and seven unit tests and documents correctness and public-key lookup attack limits of byte-wise textbook RSA.

Mixed Volume Approximations | Python and C++ numerical experiments | October 2026 | Repository

- I implemented numerical versions of the framework in my master's thesis, Approximation of Mixed Volumes of Special Convex Bodies (Chapters 4–6): B1 estimates volume, B2 samples a geometric component label, and B3 estimates component probabilities. Algorithms 1–3 organize sampling, scaling-parameter search and mixed-volume estimation. I checked the cube-plus-ball estimates against exact formulas and ran matched Python/C++ pilot comparisons. Finite burn-in, correlated samples and floating-point calculations do not establish the ideal-oracle guarantees or certify mixing and accuracy.

EDUCATION

M.Sc., Computer and System Sciences | Tata Institute of Fundamental Research, School of Technology and Computer Science, Mumbai | 2021-2026 | First Class

M.Sc., Mathematics | Tata Institute of Fundamental Research, Centre for Applicable Mathematics, Bengaluru | 2019-2021 | First Class

B.Sc. (Honours), Mathematics | Serampore College, University of Calcutta | 2015-2018 | First Class

SELECTED COURSEWORK

Computer and System Sciences: Mathematical Foundations for Systems Sciences; Algorithms; Mathematical Logic; Cryptography; Advanced Probability; Polynomials and Applications; Topics in the Study of Markov Chains (half-course); Coding Theory; Topics in High Dimensional Geometry. Audited: Convex Optimization; Random Matrix Theory.

Mathematics: Linear Algebra; Algebra; Probability; Numerical Analysis; Real and Complex Analysis; Measure and Integration; Functional Analysis; Topology; Ordinary and Partial Differential Equations.

PRESENTATIONS, ACADEMIC ACTIVITIES & RECOGNITION

- I spoke at the TIFR STCS volume-approximation seminar on 26 July 2024 and organized the Private Sequential Stateless Protocols seminar on 6 December 2024.
- Research presentation: Approximation of Mixed Volumes of Convex Bodies, STCS Symposium, TIFR, August 2025; seminar organization: Catalytic Graph Algorithms, STCS student seminar, TIFR, March 2026.
- Convex and Discrete Geometry Summer School, Alfréd Rényi Institute, Budapest, August-September 2023; School on Cluster Algebras, ICTS-TIFR, December 2018; Introduction to SageMath, ICT Mumbai and MTTS Trust, May-June 2020.
- TIFR / Department of Atomic Energy Research Scholarship, 2019-2026; DST INSPIRE Scholarship, 2015-2019.
- Qualified JAM Mathematics (2018) and Mathematical Statistics (2019); ISI M.Math admission test (2018); CMI M.Sc. Data Science written test (2019).
- Languages: Bengali, English and Hindi. References: three TIFR faculty referees available on request.